

## CUBES AND BINARY LOGIC\_25-9

**Directions for questions 1 to 4:** Select the correct alternative from the given choices.

If five cuts are made on a cube, what is the minimum number of pieces obtained?

☒ a) 18 (Your answer - ✖)

☐ b) 6

☐ c) 16

☐ d) 25

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**Solution:**

If all the five cuts are made parallel to one face, the number of pieces obtained is  $5 + 1 = 6$ .  
Choice (B)

If six cuts are made on a cube, what is the maximum number of identical pieces obtained?

☒ a) 16 (Your answer - ✖)

☐ b) 18

☐ c) 36

☐ d) 27

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**Solution:**

If two cuts are made parallel to each perpendicular plane, then the number of pieces obtained is  $3^3 = 27$ .  
Choice (D)

If two, three and four cuts are made parallel to different faces of a cube, then what is the number of identical pieces obtained?

- ☐ a) 60
- ☐ b) 30
- ☐ c) 48
- ☐ d) 24

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**Solution:**

$(2 + 1) \times (3 + 1) \times (4 + 1) = 3 \times 4 \times 5 = 60$  identical pieces are formed.  
Choice (A)

**Directions for questions 1 to 4:** Select the correct alternative from the given choices.

What is the minimum number of cuts required to cut a cube into 216 identical pieces?

- ☐ a) 36
- ☐ b) 18
- ☐ c) 15
- ☐ d) 12

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**Solution:**

$(n + 1)^3 = 216$  ( $n$  = number of cuts parallel to each face)  
 $\Rightarrow n + 1 = 6$   
 $n = 5$   
 $\therefore 3n = 15$  cuts are required.  
Choice (C)

**Q5**

**Directions for questions 5 to 7:** These questions are based on the following information.

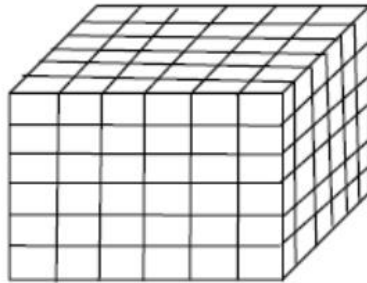
Each of 216 small identical cubes are painted blue on all faces and all these cubes are arranged to form a large cube. Now all the faces of the large cube are painted pink.

How many small cubes have only one colour on them?

- ☐ a) 96
- ☐ b) 125
- ☐ c) 64
- ☐ d) 48

**Solution:**

$$\begin{aligned}\text{Here } n^3 &= 216 \\ \Rightarrow n &= 6\end{aligned}$$



$$\text{Cubes with exactly one colour} = (n - 2)^3 = 4^3 = 64.$$

Choice (C)

**Q6**

**Directions for questions 5 to 7:** These questions are based on the following information.

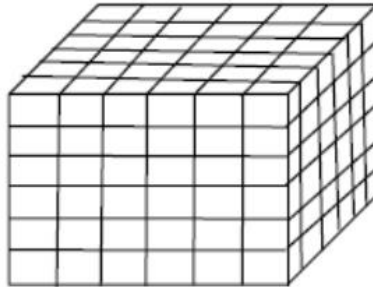
Each of 216 small identical cubes are painted blue on all faces and all these cubes are arranged to form a large cube. Now all the faces of the large cube are painted pink.

How many small cubes have exactly two faces painted pink?

- ☐ a) 36
- ☐ b) 48
- ☐ c) 64
- ☐ d) 80

**Solution:**

$$\begin{aligned}\text{Here } n^3 &= 216 \\ \Rightarrow n &= 6\end{aligned}$$



$$\text{Cubes with exactly two faces painted pink} = 12(n - 2) = 12 \times 4 = 48.$$

Choice (B)

Q7



**Directions for questions 5 to 7:** These questions are based on the following information.

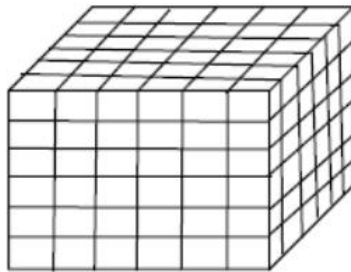
Each of 216 small identical cubes are painted blue on all faces and all these cubes are arranged to form a large cube. Now all the faces of the large cube are painted pink.

How many small cubes have exactly three faces painted blue?

- ☐ a) 8
- ☐ b) 4
- ☐ c) 2
- ☐ d) 6

**Solution:**

$$\begin{aligned}\text{Here } n^3 &= 216 \\ \Rightarrow n &= 6\end{aligned}$$



The cubes at the corners have exactly three faces painted blue i.e., 8 cubes.  
Choice (A)

Q8



**Directions for questions 8 and 9:** These questions are based on the following information.

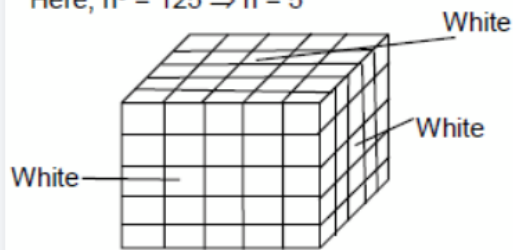
Each of 125 small identical cubes are painted black on all faces and all these cubes are arranged to form a large cube. This large cube is placed at the corner of a large room and all the visible faces of this cube are painted white.

How many smaller cubes have at least two faces with white paint?

- ☐ a) 13
- ☐ b) 16
- ☐ c) 15
- ☐ d) 10

**Solution:**

Here,  $n^3 = 125 \Rightarrow n = 5$



Exactly two faces with white paint =  $3 \times 4 = 12$  (3 edges with four cubes each)

Exactly three faces with white paint = 1

$\therefore$  Required number of cubes = 13.

Choice (A)

Q9



**Directions for questions 8 and 9:** These questions are based on the following information.

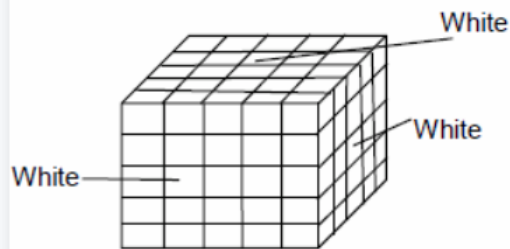
Each of 125 small identical cubes are painted black on all faces and all these cubes are arranged to form a large cube. This large cube is placed at the corner of a large room and all the visible faces of this cube are painted white.

How many smaller cubes have no face painted white?

- ☐ a) 27
- ☐ b) 100
- ☐ c) 64
- ☐ d) 81

**Solution:**

Here,  $n^3 = 125 \Rightarrow n = 5$



The required number of cubes  
 $= (5 - 1) \times (5 - 1) \times (5 - 1) = 4^3 = 64.$

Choice (C)

## Q10



**Directions for questions 10 to 13:** Select the correct alternative from the given choices.

In 'Honolulu' island, there are two types of people – truth tellers and liars. Truth-tellers always speak truth and liars always lie. I met three residents Ho, Lo and Po, and asked them "who among you is the liar?" The following are their replies.

Ho: I am a truth-teller.

Lo: Ho is not a truth-teller.

Po: Lo is not a liar.

If it is known that exactly one person among them is a liar and the other two are truth-tellers, then who among them is the liar?

- ☐ a) Ho
- ☐ b) Lo
- ☐ c) Po
- ☐ d) Cannot be determined

### Solution:

First let us take the statements made by Lo and Po. If we assume that Lo is a liar, then Po must be a liar. As we know that there is only one liar, Lo and Po cannot be the liars.

So Ho must be the liar and Lo and Po are truth-tellers.

Choice (A)

**Q11**

**Directions for questions 10 to 13:** Select the correct alternative from the given choices.

One day I met three persons – Chintu, Sintu and Mintu. Each of them belonged to a different tribe among truth-tellers, liars and alternators.

It is also known that the truth-teller always speaks truth, the liar always lies and the alternator always alternates between truth and lie in any order.

When I asked to introduce themselves, each of them made two statements, as given below.

Chintu: I am a truth-teller.

Sintu is the liar.

Sintu: I am not a liar.

Mintu is not a truth-teller.

Mintu: I am a truth-teller.

Chintu is an alternator.

Who among them is the liar?

- ☐ a) Chintu
- ☐ b) Sintu
- ☐ c) Mintu
- ☐ d) Cannot be determined

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**Solution:**

First let us assume that Chintu is a truth-teller. Then, Sintu will be the Liar. If so, according to Sintu's second statement. Mintu must be the Truth-teller, which is not possible.

Now, let us assume that Sintu is a truth-teller. So, the statements made by Chintu are false. Then, the statements of Mintu are also false, But their is not possible.

Now, if Mintu is a truth-teller, then from the second statement of Mintu we get that Chintu is an alternator. So, one statement of Chintu must be true and other must be false. As the first statement made by Chintu is false, second statement must be true, i.e. Sintu is the liar, which satisfies the statements made by Sintu.

So, Sintu is the Liar.  
It can be represented in a tabular form.

| Name   | Statement<br>I | Statement<br>II | Tribe        |
|--------|----------------|-----------------|--------------|
| Chintu | F              | T               | Alternator   |
| Sintu  | F              | F               | Liar         |
| Mintu  | T              | T               | Truth-teller |

Choice (B)

Three friends – Aooloo, Booloo, Cooloo were returning from their school. They were counting the number of pens with each of them. We know that, they together have a total of six pens and no two of them have the same number of pens. Each of them belong to different tribes among truth-tellers – who always speak truth, liars – who always speak lie and alternators – who always alternate between truth and lie in any order. They made the following statements.

Aooloo: I have more number of pens than Cooloo. Booloo has only one pen.

Booloo: I have only one pen.

Cooloo has three pens.

Cooloo: I have two pens.

Booloo has three pens.

Who among them has the maximum number of pens?

- ☐ a) Aooloo
- ☐ b) Booloo
- ☐ c) Cooloo
- ☐ d) Either (B) or (C)

### Solution:

Let each person be denoted by the first letter of their names. Here the second statement of A and the first statement of B are the same. So either of them must be true or false. Let us assume that, the statements are false. As A and B made at least one false statement, C must be the truth-teller.

From the statements of C (considered as true) we get that \_\_\_\_

(1) C has two pens.

(2) B has three pens.

∴ A has one pen.

So we can say that the first statement of A, i.e.

A has more number of pens than C is false.

Also the second statement of B, i.e. C has three pens is false.

⇒ Both A and B are liars.

So, our assumption is wrong.

∴ The second statement of A and the first statement of B (ie. B has one pen) are true.

As A and B has made at least a true statement, C must be the liar. As, both the statements of C are false, his first statement, (i.e. he has two pens), is wrong.

So, he has three pens.

∴ The second statement of B is true.

So, the first statement of A is false.

| Name | Statement I | Statement II | Tribe          | Pens |
|------|-------------|--------------|----------------|------|
| A    | F           | T            | Alternator     | 2    |
| B    | T           | T            | Truth - teller | 1    |
| C    | F           | F            | Liar           | 3    |

∴ Cooloo has the maximum number of pens.

Choice (C)

### Q13



**Directions for questions 10 to 13:** Select the correct alternative from the given choices.

Manju, Anju and Ranju, were asked who is the tallest among them. Each of them made two statements and no two of them made the same number of true statements. The statements made by them are as follows.

Manju: I am the tallest.

Anju is the shortest.

Anju: I am the tallest.

Ranju is the shortest.

Ranju: I am not the tallest.

Manju is not the tallest.

If it is known that no two of them are of the same height, then who among them had definitely made at least one true statement?

- ☐ a) Manju
- ☐ b) Anju
- ☐ c) Ranju
- ☐ d) Cannot be determined

**Solution:**

In this question, as they did not make the same number of true statements, we can say that one of them made two true statements, one of them made one true statement and other one made no true statement.

First, let us assume both the statements of Manju are true, i.e., Manju is the tallest and Anju is the shortest. So Ranju is the second tallest. So, the first statement of Anju i.e. 'I am the tallest', is not true. The second statement of Anju, i.e. 'Ranju is the shortest', is not true. As both the statements of Anju are false, one statement of Ranju must be true. The first statement of Ranju, i.e. 'I am not the tallest', is true. So, the second statement of Ranju must be false, i.e. Manju is not that tallest, which satisfies the conditions.

The result can be represented in a tabular form.

**Table 1**

| Name  | Statement I | Statement II | Height (rank) |
|-------|-------------|--------------|---------------|
| Manju | T           | T            | 1             |
| Anju  | F           | F            | 3             |
| Ranju | T           | F            | 2             |

Now, let us assume that both the statements of Anju are true. From the statements of Anju we get that Anju is the tallest and Ranju is the shortest. So, Manju is the second tallest.

The first statement of Ranju, i.e. 'I am not the tallest' is true. The second statement of Ranju, i.e. Manju is not the tallest, is also true, which is not possible.

Now, let us assume that both the statements made by Ranju are true. From the statements made by Ranju we get that Anju is the tallest. So, the first statement of Anju is true. So, the second statement of Anju is false. So, Ranju is the second tallest and Manju is the shortest. Now both the statements made by Manju are false, which satisfies the conditions.

The result can be represented in a tabular form.

**Table – 2**

| Name  | Statement I | Statement II | Height (rank) |
|-------|-------------|--------------|---------------|
| Manju | F           | F            | 3             |
| Anju  | T           | F            | 1             |
| Ranju | T           | T            | 2             |

Comparing tables (1) and (2) we get that one statement of Ranju is definitely true.  
Choice (C)



**Directions for questions 14 and 15:** These questions are based on the following information.

One day I met three persons A, B and C. Each of them has an umbrella and a bag. The three umbrellas are of different colours among green, yellow and red, and the three bags are of different colours among green, yellow and red. None of them has both the items of same colour. When I asked them about the colour of their bag and umbrella, each of them made two statements as follows.

A: I have a green umbrella and a yellow bag.

B has a green umbrella or a red bag.

B: I have a red umbrella and a green bag.

C has a yellow umbrella or a green bag.

C: I have a yellow bag and a green umbrella.

A has neither a green umbrella nor a yellow bag.

It is also known that one of them always speaks truth, one always lies and the other person alternates between truth and lie in any order.

If at least one of the statements of C is true, then who has the green bag?

- ☐ a) A
- ☐ b) B
- ☐ c) C
- ☐ d) Cannot be determined

### Solution:

Assuming both the statements of A as true.

Table (i)

| Name | I | II | Umbrella | Bag |
|------|---|----|----------|-----|
| A    | T | T  | G        | Y   |
| B    | F | T  | Y        | R   |
| C    | F | F  | R        | G   |

Assuming both the statements of B as true.

Table – (ii)

| Name | I | II | Umbrella | Bag |
|------|---|----|----------|-----|
| A    | T | F  | G        | Y   |
| B    | T | T  | R        | G   |
| C    | F | F  | Y        | R   |

Assuming both the statement of C as true.

Table (iii)

| Name | I | II | Umbrella | Bag |
|------|---|----|----------|-----|
| A    | F | T  | R        | G   |
| B    | F | F  | Y        | R   |
| C    | T | T  | G        | Y   |

Or

Table (iv)

| Name | I | II | Umbrella | Bag |
|------|---|----|----------|-----|
| A    | F | F  | Y        | R   |
| B    | T | F  | R        | G   |
| C    | T | T  | G        | Y   |

Table (iv)

| Name | I | II | Umbrella | Bag |
|------|---|----|----------|-----|
| A    | F | F  | Y        | R   |
| B    | T | F  | R        | G   |
| C    | T | T  | G        | Y   |

In the case of both the statements of C are true either A or B has the green bag.  
[from tables (iii) and (iv)]

Choice (D)

If at least one of the statements made by C is a lie, then who among them has the green umbrella and the yellow bag?

- ☐ a) A
- ☐ b) B
- ☐ c) C
- ☐ d) Cannot be determined

**Solution:**

Assuming both the statements of A as true.

Table (i)

| Name | I | II | Umbrella | Bag |
|------|---|----|----------|-----|
| A    | T | T  | G        | Y   |
| B    | F | T  | Y        | R   |
| C    | F | F  | R        | G   |

Assuming both the statements of B as true.

Table – (ii)

| Name | I | II | Umbrella | Bag |
|------|---|----|----------|-----|
| A    | T | F  | G        | Y   |
| B    | T | T  | R        | G   |
| C    | F | F  | Y        | R   |

Assuming both the statement of C as true.

Table (iii)

| Name | I | II | Umbrella | Bag |
|------|---|----|----------|-----|
| A    | F | T  | R        | G   |
| B    | F | F  | Y        | R   |
| C    | T | T  | G        | Y   |

Or

Table (iv)

| Name | I | II | Umbrella | Bag |
|------|---|----|----------|-----|
| A    | F | F  | Y        | R   |
| B    | T | F  | R        | G   |
| C    | T | T  | G        | Y   |

Or

Table (iv)

| Name | I | II | Umbrella | Bag |
|------|---|----|----------|-----|
| A    | F | F  | Y        | R   |
| B    | T | F  | R        | G   |
| C    | T | T  | G        | Y   |

This satisfies the conditions of table (i) and (ii).  
In both the tables A has the green umbrella and the yellow bag.

Choice (A)